Alec Cambre

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EDUCATION:

Duke University Aug 2026-Dec 2027
Master of Engineering, Mechanical Engineering

Southern Methodist University Aug 2022-May 2026
Bachelor of Science, Mechanical Engineering | Bachelor of Arts, Chemistry

- GPA: 3.71

RESEARCH & WORK EXPERIENCE

Duke University – Bridgeman Lab Durham, NC
Graduate Researcher in Surgical Robotics Aug 2026-Present

- Validating MPC-controlled, 6-DOF arm utilizing an OTC-guided laser ablator for autonomous brain tumor removal
- Consolidating GitHub codebase and authoring an onboarding guide for robotic control, code annotation, and execution
- Fabricating multi-layered phantom tumors to validate MPC simulation results and supporting controller optimization

Southern Methodist University - Templeton Chair Lab Dallas, TX
Undergraduate Researcher May 2025-Dec 2025

- Designed electrospinning apparatus to increase surface area and sensitivity of electrode for biosensing of antibodies
- Conducted comparative analysis for developing nanostructures considering polymer reactions, EHDP, and etching
- Built the functional rig, creating a system that creates a microstructure mesh of around 7 cm in diameter on a receiver
- Conducted initial optimizations considering flow rate, flight distance, and voltage to reduce fiber diameter to < 1 um
- Generated strategies for localized dissolution and Au doping to meet conductivity and prevent shorting for future work

ARCO/Murray Dallas, TX
Project Management Intern May 2024-Aug 2024

- Managed 2 construction sites by overseeing timelines and coordination, crafting budgets, and negotiating contracts
- Formulated a $9,000,000+ competitive technical bid, won by ARCO/Murray, by auditing sub cost and tuning scope
- Designed value engineering alts of $120,000+ in Bluebeam Revu by evaluating local IBC code for overdesigns
- Negotiated with subcontractors on competitive bid pricing, saving near 40% on costs and ensuring quality and coverage
- Organized on-site QC walks and punch lists using Procore to verify installations and ensure OSHA compliance

PROJECTS

Wireless Power for Heart Implant Aug 2025-May 2026

- Engineered a wireless inductive LVAD prototype with < 5 cm coupled coils to increase survivability and affordability
- Constructed a heart-mimicking system to test if power output drove 2.1 psi blood pressure and 6 L/min flow on output
- Evaluated pumps and suppliers against power and size requirements to identify a suitable option within a $400 budget
- Optimized circuit design to minimize energy absorption into tissue, ensuring power delivery without cooking muscle
- Adapted design to conform to IEEE, FDA, & FCC regulations and standards for the design of wireless medical devices

Shock Analysis for Tennis Racket - Experimental Design Nov 2025-Dec 2025

- Created Arduino-based experimental apparatus testing shock in tennis rackets via custom accelerometer configuration
- Engineered a DAQ module data streaming to MS Excel, tuned by altering sampling frequency and sensor sensitivities
- Analyzed over 40,000+ data points using statistical methods to quantify vibration data and evaluate test consistency

Drone Assembly Design | FEA & CFD Analysis Oct 2025-Dec 2025

- Modeled a racing drone assembly in SolidWorks CAD using manufacturer specs and adhering to competition rules
- Performed FEA to evaluate stresses, deformation, and heat transfer, verifying performance under expected racing loads
- Interpreted CFD simulations to visualize rotor-induced airflow and analyze lift-generating mechanisms across assembly

Optimization of Fin System for Heat Dissipation Feb 2025-May 2025

- Engineered an air-cooled fin system for a 12HP internal combustion engine, achieving 4.186 kW heat dissipation
- Determined optimal fin parameters that maximize airflow dissipation using 25,000+ data points via Excel models
- Recommended configurations based on design manufacturability/efficiency confirmed via SolidWorks modelling

ORGANIZATIONS & LEADERSHIP

Delta Sigma Phi, Lambda Chapter Dallas, TX
Founding Vice President of Recruitment Dec 2023-Jan 2025

- Recruited 45 members across 2 terms, expanding chapter membership 125%+ and managing outreach to 1000+ leads
- Utilized a budget of $70,000 efficiently, creating high retention recruitment campaigns while saving $10,000+ total
- Coordinated recruitment across a 5-member team, establishing outreach cadence, event logistics, and lead procurement

TECHNICAL SKILLS & INTERESTS

Skills: MATLAB, SolidWorks, Soldering, FEA, CFD, Simulink, Machining, GitHub, Python (Beginner), ROS 2, MS 365
Interests: Chess, Bartending, Surgical Robotics, Drones, Soccer, Dallas Mavericks, Eagle Scout, SMU Football